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Research paper

Longitudinal characteristics of lymphocyte responses and cytokine profiles in the peripheral blood of SARS-CoV-2 infected patients

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ABSTRACT

Background: The dynamic changes of lymphocyte subsets and cytokines profiles of patients with novel coronavirus disease (COVID-19) and their correlation with the disease severity remain unclear.
Methods: Peripheral blood samples were longitudinally collected from 40 confirmed COVID-19 patients and examined for lymphocyte subsets by flow cytometry and cytokine profiles by specific immunoassays.
Findings: Of the 40 COVID-19 patients enrolled, 13 severe cases showed significant and sustained decreases in lymphocyte counts [0·6 (0·6-0·8)] but increases in neutrophil counts [4·7 (3·6-5·8)] than 27 mild cases [1.1 (0·8-1·4); 2·0 (1·5-2·9)]. Further analysis demonstrated significant decreases in the counts of T cells, especially $CD8^+$ T cells, as well as increases in IL-6, IL-10, IL-2 and IFN-γ levels in the peripheral blood in the severe cases compared to those in the mild cases. T cell counts and cytokine levels in severe COVID-19 patients who survived the disease gradually recovered at later time points to levels that were comparable to those of the mild cases. Moreover, the neutrophil-to-lymphocyte ratio (NLR) (AUC=0·93) and neutrophil-to-$CD8^+$ T cell ratio (N8R) (AUC =0·94) were identified as powerful prognostic factors affecting the prognosis for severe COVID-19.
Interpretation: The degree of lymphopenia and a proinflammatory cytokine storm is higher in severe COVID-19 patients than in mild cases, and is associated with the disease severity. N8R and NLR may serve as a useful prognostic factor for early identification of severe COVID-19 cases.
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1. Introduction

First reported in Wuhan, China, on 31 December 2019, an ongoing outbreak of a viral pneumonia in humans has raised acute and grave global concern, and rapidly spread to 197 countries, areas or territories [1]. The causative pathogen was rapidly identified as a novel

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Research in context

Evidence before this study

Lymphopenia and inflammatory cytokine storm have been shown in severe acute respiratory syndrome coronavirus (SARS-CoV), the Middle East respiratory syndrome coronavirus (MERS-CoV) infections, and coronavirus disease (COVID-19).

Added value of this study

For the first time, the kinetic changes of lymphopenia, lymphocyte subset and cytokine profile were longitudinally characterized. The significant decreases in the count of $CD8^{+}$T cell number and increase in inflammatory cytokine levels (e.g., IL-6, IL-10) are dynamically correlated with the severity of COVID-19 patients. T cell counts and cytokine levels in severe COVID-19 patients who survived the disease gradually recovered at later time points to levels that were comparable to those of the mild cases. Neutrophil-to-lymphocyte ratio (NLR) (AUC=0·93) and neutrophil-to- $CD8^{+}$ T cell ratio (N8R) (AUC =0·94) were identified as powerful prognostic factors affecting the prognosis for severe COVID-19.

Implications of all the available evidence

The development of severe COVID-19 is the result of imbalanced inflammation and antiviral immune responses. N8R and NLR can serve as useful prognostic factor for early identification of severe COVID-19 cases. This will help physicians to provide timely intervention for COVID-19.

β-coronavirus, which has since been formally named as the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by the International Committee on Taxonomy of Viruses. According to the World Health Organization coronavirus disease 2019 (COVID-19) situation report and the daily report of the National Health Commission of China, the epidemic of SARS-CoV-2 has so far caused 462,684 confirmed cases (81,968 cases in China, including 11,977 severe cases), and 20,834 deaths (3293 deaths in China) in the world by March 26th, 2020 [1,2]. The disease caused by SARS-CoV-2 has been recently named as the COVID-19 by the World Health Organization. Previous studies about the epidemiological and clinical characteristics of COVID-19 have shown that patients with COVID-19 may develop either mild or severe symptoms of acute respiratory infection, with the mild case patients showing symptoms of fever, dry cough, fatigue, abnormal chest CT findings but with a good prognosis [3,4]. In contrast, some patients develop severe pneumonia, acute respiratory distress syndrome (ARDS) or multiple organ failure, with death rates ranging from between 4·3% to 15% according to different study reports [3,5].

Lymphopenia (lymphocyte count $<1{\cdot}0 \times 10^{9}$/L)[3] and inflammatory cytokine storm are typical laboratory abnormalities observed during highly pathogenic coronavirus infections, such as the severe acute respiratory syndrome coronavirus (SARS-CoV) and the Middle East respiratory syndrome coronavirus (MERS-CoV) infections, and are believed to be associated with disease severities [6,7]. Recent studies have also reported decreases in the counts of lymphocytes (e.g., $CD4^{+}$ T cell, $CD8^{+}$ T cell) in the peripheral blood and increases in serum inflammatory cytokine levels (e.g., IL-6) in COVID-19 patients [5,8–10]. However, it has remained largely unclear in the kinetics of lymphocyte subsets and inflammatory cytokines change in the peripheral blood during COVID-19. In this study, we longitudinally characterized the changes of lymphocyte subsets and cytokines profiles in the peripheral blood of COVID-19 patients with distinct disease severities.

2. Methods

2.1. Data collection

A written informed consent was regularly obtained from all patients upon admission into Wuhan Union Hospital, China. The study was approved by the Ethics Committee of Wuhan Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, in China. The 40 confirmed COVID-19 patients at Wuhan Union Hospital during January 5 to January 24, 2020 were enrolled into this retrospective single-center study. All medical record information including epidemiological, demographic, clinical manifestation, laboratory data, and outcome data were obtained. All data were checked by a team of trained physicians.

2.2. Laboratory examination

Laboratory confirmation of the SARS-CoV-2 was performed by local CDC according to Chinese CDC protocol. Throat-swab specimens were collected from all patients and the samples were maintained in a viral-transport medium for laboratory testing. An infection with other respiratory viruses including influenza A virus, influenza B virus, coxsackie virus, respiratory syncytial virus, parainfluenza virus and enterovirus was excluded by real-time RT-PCR. Specimens, including sputum or bronchoalveolar lavage fluid, blood, urine, and feces, were cultured to identify pathogenic bacteria or fungi that may be associated with the SARS-CoV-2 infection. The specific IgG and IgM of chlamydia pneumonia and mycoplasma pneumonia were detected by chemiluminescence immunoassay. The lymphocyte test kit (Beckman Coulter Inc., FL, USA) was used for lymphocyte subset analysis. Plasma cytokines (IL2, IL4, IL6, IL10, TNF - α and IFN - γ) were detected using the human Th1/2 cytokine kit II (BD Ltd., Franklin lakes, NJ, USA). All tests were performed according to the product manual.

2.3. Statistical analyses

Classification variables are expressed in frequency or percentage, and significance was detected by chi square or Fisher's exact test. The quantized variables of parameters are expressed as mean ± standard deviation, and the significance is tested by *t*-test. Nonparametric variables are expressed in median and quartile intervals, and significance was tested by Mann Whitney U or Kruskal Wallis test. Data (nonnormal distribution) from repeated measures were compared using the generalized linear mixed model. $P < 0{\cdot}05$ was considered statistically significant in all statistical analyses. Principal component analysis (PCA) was performed to identify the major contributing factors among clinical parameters to distinguish between mild and severe cases of COVID-19 patients. The diagnostic values of selected parameters for differentiating between mild and severe cases of COVID-19 patients were assessed by receiver operating characteristic (ROC) and the area under the ROC curve (AUC). SPSS statistical software (Macintosh version 26·0, IBM, Armonk, NY, USA) and R package were used for statistical analysis.

3. Results

3.1. Demographic and clinical characteristics of COVID-19 patients

The diagnosis of COVID-19 for patients was performed according to the Guidelines of the Diagnosis and Treatment of New Coronavirus Pneumonia (version 5) published by the National Health Commission of China. Mild patients met all following conditions: (1) Epidemiological history, (2) Fever or other respiratory symptoms, (3) Typical CT image abnormities of viral pneumonia, and (4) Positive result of RT-PCR for SARS-CoV-2 RNA. Severe patients additionally met at least

Table 1
Demographics and baseline characteristics of patients infected with SARS-CoV-2.

Baseline variables	All patients (*N* = 40)	Mild patients (*N* = 27)	Severe patients (*N* = 13)
Characteristics			
Age (year)	48·7 ± 13·9	43·2 ± 12·3	59·7 ± 10·1
Gender (%)			
Men	15 (37·5)	8 (29·6)	7 (53·8)
Women	25 (62·5)	19 (70·4)	6 (46·2)
Huanan seafood market exposure (%)	3 (7·5)	1 (3·7)	2 (5·4)
Current smoking	5 (12·5)	3 (11·1)	2 (15·4)
Underlying diseases (%)	14 (35·0)	7 (25·9)	7 (53·8)
Diabetes	6 (15·0)	2 (7·4)	4 (30·8)
Hypertension	6 (15·0)	1 (3·7)	5 (38·5)
Pituitary adenoma	2 (5·0)	1 (3·7)	1 (7·7)
Thyroid disease	2 (5·0)	2 (7·4)	0
Malignancy	2 (5·0)	2 (7·4)	0
Co-infection (%)	5 (12·5)	0	5 (38·5)
Fungi	4 (10·0)	0	4 (30·8)
Bacteria	1 (2·5)	0	1 (7·7)
Signs and symptoms			
Fever	36 (90·0)	23 (85·2)	13 (100)
Highest temperature, °C			
<37.3	4 (10·0)	4 (14·8)	0
37.3–38.0	10 (25·0)	8 (29·6)	2 (15·4)
38.1–39.0	17 (42·5)	9 (33·3)	8 (61·5)
>39.0	9 (22·5)	6 (22·2)	3 (23·1)
Chill	10 (25·0)	5 (18·5)	5 (38·5)
Shivering	5 (12·5)	2 (7·4)	3 (23·1)
Fatigue	22 (55·0)	14 (51·9)	8 (61·5)
Cough	33 (82·5)	22 (81·5)	11 (84·6)
Sputum production	21 (52·5)	11 (40·7)	10 (76·9)
Pharyngalgia	5 (12·5)	4 (14·8)	1 (7·7)
Dizziness	7 (17·5)	4 (14·8)	3 (23·1)
Headache	8 (20·0)	6 (22·2)	2 (15·4)
Rhinorrhea	1 (2·5)	1 (3·7)	0
Chest tightness	12 (30·0)	7 (25·9)	5 (38·5)
Chest pain	1 (2·5)	1 (3·7)	0
shortness of breath	5 (12·5)	5 (18·5)	0
Dyspnea	1 (2·5)	1 (3·7)	0
Myalgia	15 (37·5)	7 (25·9)	8 (61·5)
abdominal pain	1 (2·5)	1 (3·7)	0
Diarrhea	3 (7·5)	1 (3·7)	2 (15·4)
Nausea	3 (7·5)	0	3 (23·1)
Vomiting	1 (2·5)	0	1 (7·7)
Hypoleucocytosis	10 (25·0)	8 (29·6)	2 (15·4)
Lymphopenia	23 (57·5)	12 (44·4)	11 (84·6)
Thrombocytopenia	5 (12·5)	3 (11·1)	2 (15·4)
Duration of hospitalization (day)	12·6 ± 6·7	12·9 ± 7·1	11·8 ± 5·9
ARDS	3 (7·5)	0	3 (23·1)
prognosis			
Hospitalization	4 (10)	1 (3·7)	3 (23·1)
Discharge	33 (82·5)	26 (96·3)	7 (53·8)
Death	3 (7·5)	0	3 (23·1)

one of the following conditions: (1) Shortness of breath, respiratory rate ≥30 times/min, (2) Oxygen saturation (Resting state) ≤93%, or (3) PaO2 / FiO2 ≤300 mmHg. A total of 40 patients were enrolled in this study, which were all Wuhan residents and laboratory confirmed cases. The flowchart of patient enrollment is shown in Fig. S5. The patients were divided into two groups according to above-mentioned conditions, including 27 mild cases (67·5%) and 13 severe cases (32·5%). Three patients in the severe group died on day 15, 18 and 21 after disease onset.

The enrolled COVID-19 patients consisted of 15 males (37·5%) and 25 females (62·5%) (Table 1). Only 3 patients (7·5%) had an exposure history (shopping) to the Huanan seafood market in Wuhan. The medium age of the patients was 48·7 ± 13·9 years old. The ages of the severe patient group (59·7 ± 10·1 years) were older than that of the mild group (43·2 ± 12·3 years). The duration of hospitalization was 12·6 ± 6·7 day, and three [7·5%] of the patients died. A total of 14 (35%) patients in both groups had underlying chronic medical conditions, including diabetes (6 [15%]), hypertension (6 [15%]), pituitary adenoma (2 [5%]), thyroid disease (2 [5%]) and tumor disease (2 [5%]). Four severe patients had mixed fungal infection and 1 severe patient had a mixed bacterial infection (Table 1). All severe patients and 85·2% of the mild patients had fever, while no significant difference in the degrees of temperature was observed between the two groups (Table 1). The severe patients showed significantly higher frequencies in the occurrence of sputum production, myalgia and nausea. Three patients in the severe group developed (acute respiratory distress syndrome, ARDS). All patients received antiviral treatment, including interferon α2b (16, 40·0%), ribavirin (24, 60·0%), abidol (18, 45·0%), and/or oseltamivir (17, 42·5%), as well as antibiotic treatment, including moxifloxacin (40, 100%), cephalosporins (20, 50%), penicillin (1, 2·5%), and/or other antibiotics (29, 72·5%). Seventeen (42·5%) patients received antifungal treatment (13 cases took prophylactic treatment) and 8 (20%) received methylprednisolone therapy (1 mild case and 7 severe cases) (Table S1). The levels of fibrinogen, D-dimer, total bilirubin, aspartate transaminase, alanine transaminase, lactate dehydrogenase, creatine kinase, C-reactive protein (CRP), ferritin and serum amyloid A protein (SAA) in the peripheral blood of the severe patients were significantly higher at admission compared to the mild

Table 2
Comparison of laboratory parameters between mild and severe COVID-19 patients.

Baseline variables	All patients (N = 40)	Mild patients (N = 27)	Severe patients (N = 13)
Hemoglobin (g/l)	126·4 ± 13·4	127·8 ± 13·1	123·4 ± 14·0
Platelet (× 10^9/L)	183·1 ± 69·0	181·4 ± 70·7	186·6 ± 68·1
White blood cell (× 10^9/L)	4·8 ± 2·6	3·9 ± 1·5	6·6 ± 3·4
Neutrophil (× 10^9/L)	2·8 (1·6–4·3)	2·0 (1·5–2·9)	4·7 (3·6–5·8)
Lymphocyte (× 10^9/L)	0·9 (0·7–1·3)	1.1 (0·8–1·4)	0·6 (0·6–0·8)
Monocyte (× 10^9/L)	0·3 (0·2–0·5)	0·3 (0·2–0·5)	0·2 (0·2–0·5)
TBil (umol/l)	10·3 ± 5·0	8·8 ± 4·1	13·2 ± 5·5
ALT (U/L)	22·5 (16·8–31·2)	19·0 (13·5–26·0)	27·0 (23·0–50·0)
AST (U/L)	34·1 ± 17·7	25·9 ± 9·5	51·2 ± 18·7
LDH (U/L)	303·9 ± 168·8	221·5 ± 71·2	462·4 ± 190·6
CK (U/L)	59·5 (45·0–88·8)	51·0 (45·0–68·0)	104·0 (77·0–124·0)
Blood urea nitrogen (mmol/l)	3·2 (2·5–4·3)	3·2 (2·5–4·4)	3·3 (2·7–3·7)
Serum creatinine (umol/l)	67·3 ± 19·7	64·0 ± 13·3	74·2 ± 28·3
Blood potassium (mmol/l)	3·8 ± 0·5	3·9 ± 0·5	3·7 ± 0·4
Blood sodium (mmol/l)	145·9 ± 43·4	149·5 ± 52·5	138·6 ± 6·2
D-Dimer (mg/l)	0·6 (0·3–0·9)	0·4 (0·2–0·8)	0·9 (0·7–1·5)
PT (s)	13·2 ± 0·6	13·1 ± 0·6	13·4 ± 0·6
APTT (s)	39·5 ± 4·5	39·5 ± 4·6	39·5 ± 4·2
INR	1·0 ± 0·1	1·0 ± 0·1	1·0 ± 0·1
FIB (g/l)	5·1 ± 1·6	4·5 ± 1·4	6·3 ± 1·3
IgE	43·9 (14·4–98·0)	26·5 (12·8–76·1)	43·9 (27·0–105·5)
IgG	11·1 ± 2·0	10·8 ± 2·0	11·5 ± 2·0
IgA	2·2 ± 0·7	2·2 ± 0·8	2·4 ± 0·6
IgM	1·1 ± 0·4	1·1 ± 0·5	1·1 ± 0·3
C-reactive protein (mg/l)	38·1 (4·7–65·2)	7·6 (3·1–57·3)	62·9 (42·4–86·6)
Ferritin (ug/l)	596·5 (308·6–1087·6)	367·8 (174·7–522·0)	835·5 (635·4–1538·8)
SAA (mg/l)	134·4 (35·7–586·3)	46·9 (20·5–134·4)	607·1 (381·9–686·2)
C3 (g/l)	0·8 ± 0·2	0·8 ± 0·2	0·8 ± 0·1
C4 (g/l)	0·3 ± 0·1	0·3 ± 0·1	0·3 ± 0·1

patients (Table 2). No significant differences in the serum levels of immunoglobulins (IgA, IgG and IgM), complement C3 or C4 were observed between the two groups (Table 2). We also analyzed the impact of methylprednisolone treatment on the lymphocyte and neutrophil counts, as well as other cytokine levels. The results showed that patients with methylprednisolone treatment had higher lymphocyte counts, lower neutrophil counts, NLR, N8R and serum IL-6 levels than patients without methylprednisolone treatment. However, the differences were not statistically significant (Table S2).

3.2. Kinetic analysis of lymphocyte subsets in the peripheral blood of COVID-19 patients

Lymphopenia was observed in 44·4% (12/27) of mild patients and 84·6% (11/13) of severe patients at the onset of the disease. As shown in Table 2, the absolute counts of lymphocytes in the peripheral blood of the severe patients was significantly lower, while the absolute counts of total white blood cells (WBCs) and neutrophils were significantly higher than those of the mild patients at the time of hospital admission. No significant difference in monocyte counts was observed between the two groups (Table 2). Next, we analyzed the kinetic changes of WBCs, neutrophils and monocytes as well as different lymphocyte subsets in the peripheral blood of COVID-19 patients from the disease onset to at least 16 days later. The three mortalities in the severe group were excluded from the analysis due to the lack of kinetic data. Significant increases in total WBCs counts in the severe group were only observed at the time point of onset (within 3 days) but not during the following period of disease progression compared to the mild group (Fig. 1a). Significant increases in the neutrophil counts of the severe group were observed not only at the time point of disease onset, but also at 13–15 days later compared to the mild group (Fig. 1b). In contrast, a sustained decrease in the lymphocyte counts of the severe group was observed compared to those of the mild patients. The difference was significant at the time point of disease onset and became even greater on 4–6 days later (Fig. 1c). From 7 to 15 days after disease onset, the lymphocyte counts gradually increased in the severe group, and reached a comparable level to that of the mild patients at 16 days after disease onset (Fig. 1c). No significant differences in monocyte counts were observed between the two groups during the whole observation period (Fig. 1d). There were 4 patients co-infected with fungi in the severe case group. To examine the impact of fungal infection on the results, we performed the analyses by excluding the data of the 4 cases with fungal co-infections from the severe patient group, and similar kinetic changes of different subsets of PBMCs were observed when these 4 cases were excluded. Total WBCs counts in the severe group significantly increased at the time point of onset (within 3 days) but not during the following period of disease progression compared to the mild group (Fig. S4a). Significant increases in the neutrophil counts of the severe group were observed only at the time point of disease onset (Fig. S4b). In contrast, significant decreases in lymphocyte counts of the severe group were observed at the time point of disease onset and became even greater on 4–6 days later compared to those of the mild patients (Fig. S4c). No significant differences in monocyte counts were observed between the two groups during the whole observation period (Fig. S4d).

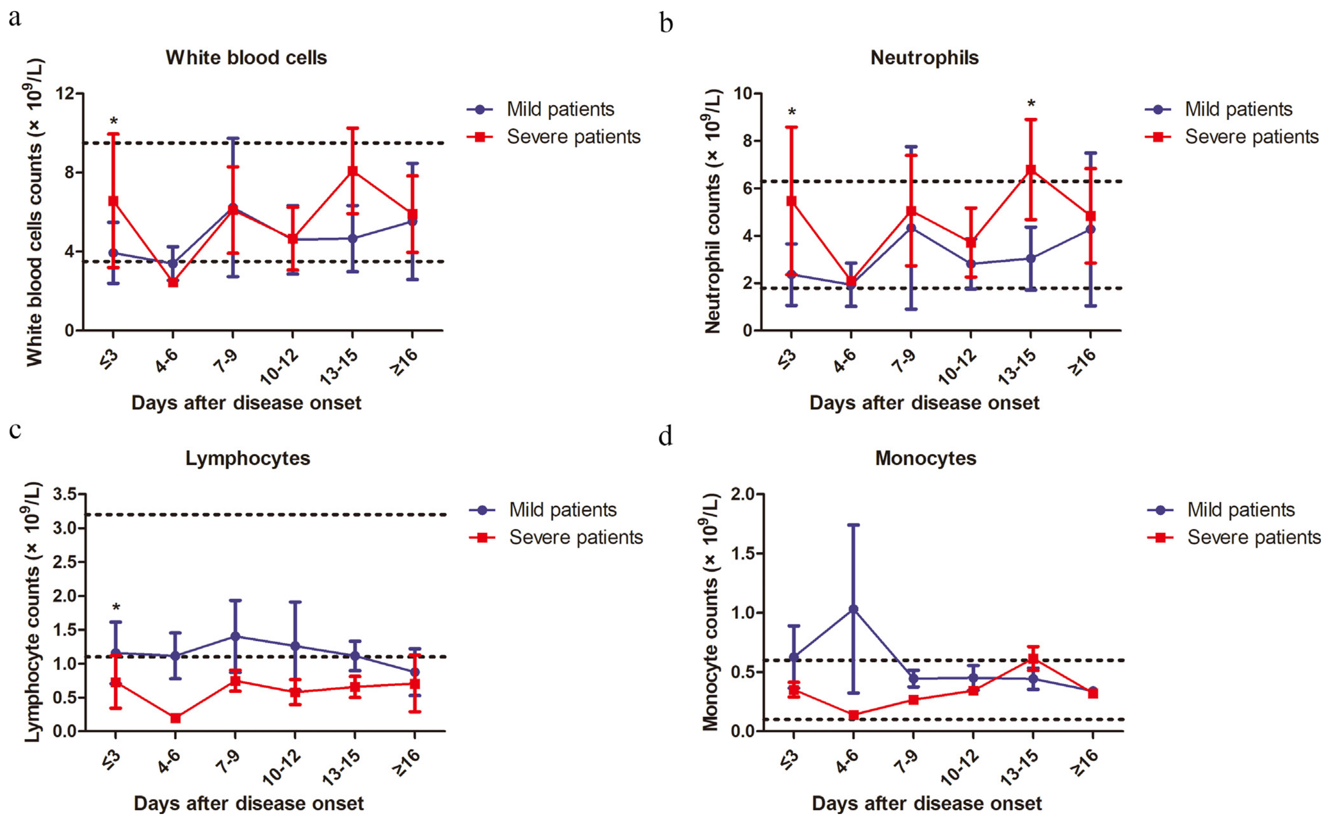

Fig. 1. Kinetic analysis of cell counts of different populations of WBCs in COVID-19 patients. The absolute numbers of total WBCs (a), neutrophils (b), lymphocytes (c) and monocytes (d) in the peripheral blood of mild (blue line) and severe (red line) COVID-19 patients were analyzed at different time points after hospital admission. Error bars, mean ± SD.; Statistics, repeated measures (mixed model) ANOVA. $^{*}p<0{\cdot}05$. The upper dotted lines show the upper normal limit of each parameter, and the lower dotted lines show the lower normal limit of each parameter. Number of cases for each time point, M: Mild patients, S: Severe patients. Time≤3d 27(M),13(S); 4-6d 10(M), 1(S); 7-9d 9 (M), 4 (S); 10-12d 6 (M), 3(S); 13-15d 4 (M), 6 (S); ≥16d 2 (M), 7 (S).

In order to further determine the kinetic changes of different lymphocyte subsets in the peripheral blood of COVID-19 patients, we performed flow cytometry to stain CD3^{+} T cells, CD4^{+} and CD8^{+} T cell subsets, B cells and NK cells. Similar to the findings for lymphocytes, sustained decreases in CD3^{+}, CD8^{+} and CD4^{+} T cell counts were observed in the severe group compared to those of the mild patients during clinical observation (Figs. 2a-c and S1). The lowest CD3^{+}, CD4^{+} and CD8^{+} T cell counts were observed at 4–6 days after disease onset (Fig. 2a-c). The differences in CD3^{+} and CD8^{+} T cell counts between the two groups were significant at the time point of disease onset and 7–9 days later (Fig. 2a and c). However, the differences in CD4^{+} T cell counts between the two groups did not reach a statistical significance at any time point (Fig. 2c). The T cell counts started to gradually increase in the severe group starting at 7 days after disease onset, and reached comparable levels to those in the mild patients on day 16 after disease onset (Fig. 2a-c). No significant differences in B cell and NK cell counts were observed between the two groups during the whole course of the disease (Fig. 2d and e).

3.3. Kinetic analysis of inflammatory cytokine levels in the serum of COVID-19 patients

A previous study demonstrated changes in inflammatory cytokine levels, such as IL-2, IL-7, IL-10, and TNF-α, in the serum of COVID-19 patients [3]. Therefore, we further characterized the kinetic changes of inflammatory cytokine levels, including IL-2, IL-4, IL-6, IL-10, IFN-γ and TNF-α, in the serum of our patient cohort. Fluctuations in the serum levels of these cytokines in the mild patient group were minor. In contrast, the severe patient group showed more significant fluctuations in the serum levels of these cytokines (Fig. 3). All examined cytokines, except IL-6, reached their peak levels in the serum at 3–6 days after disease onset (Fig. 3). Both IL-6 and IL-10 levels showed sustained increases in the severe group compared to the mild group (Fig. 3a and b). Reductions in serum IL-6 levels in the severe group started at 16 days after disease onset, and IL-10 levels were lowest at 13 days after disease onset (Fig. 3a and b). Significant increases in serum IL-2 and IFN-γ levels in the severe group were only observed at 4–6 days after disease onset (Fig. 3c and f). No significant differences in IL-4 and TNF-α levels were observed between the two groups during the whole course of the disease (Fig. 3d and e). All examined cytokines reached similar levels between the severe and mild patient groups at 16 days after disease onset (Fig. 3). We observed higher CRP levels in the severe group than the mild group at most time points; however, the differences were not significant (Fig. S6). Moreover, we also performed the analyses by excluding the 4 patients with fungal co-infections from the severe patient group. We observed that both IL-6 and IL-10 levels showed sustained increases in the severe group compared to the mild group (Fig. S3a and b). Significant increases in serum IFN-γ levels in the severe group were only observed at 4–6 days after disease onset (Fig. S3f). No significant differences in IL-2, IL4 and TNF-α levels were observed between the two groups during the whole course of the disease (Fig. S3c–e).

3.4. Prognostic factors for the identification of severe COVID-19 cases

Next, we examined the possibilities of using above-mentioned parameters as prognostic factors for identifying severe cases in COVID-19 patients. PCA was firstly performed by R package “factoextra” to identify correlated variables for distinguishing severe patients from mild patients (Fig. 4a). Four most contributing variables, neutrophil-to-CD8^{+} T cell ratio (N8R), neutrophil-to-lymphocyte ratio (NLR),

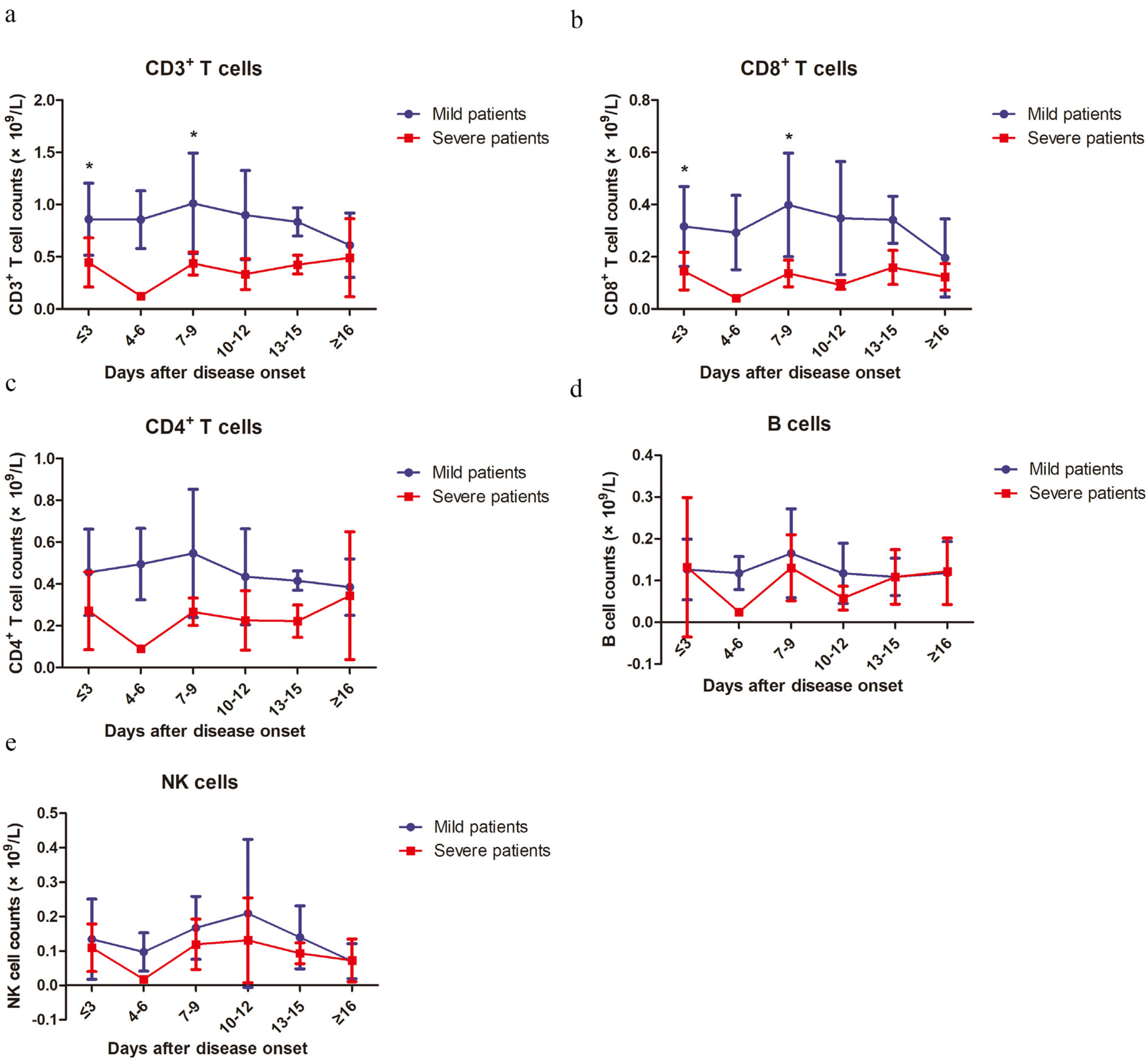

Fig. 2. Kinetic analysis of cell counts of different lymphocyte subsets in COVID-19 patients. The absolute numbers of $CD3^+$ T cells (a), $CD8^+$ T cells (b), $CD4^+$ T cells (c), B cells (d) and NK cells (e) in the peripheral blood of mild (blue line) and severe (red line) COVID-19 patients were analyzed at different time points after hospital admission. Error bars, mean ± SD.; Statistics, repeated measures (mixed model) ANOVA. $^*p<0{\cdot}05$. Number of cases for each time point, M: Mild patients, S: Severe patients. Time≤3d 27(M),13(S); 4-6d 10(M), 1 (S); 7-9d 9 (M), 4 (S); 10-12d 6 (M), 3(S); 13-15d 4 (M), 6 (S); ≥16d 2 (M), 7 (S).

neutrophil counts (NEC) and white blood cells counts (WBCC) were selected as potential prognostic factors for further detailed statistical analysis. To assess the diagnostic value of these four selected parameters, the receiver operating characteristic (ROC) curve and the area under ROC curve (AUC) were calculated by R package "pROC" (Fig. 4b). The results of this analysis identified N8R with a higher AUC (0·94) than NLR (0·93), NE (0·91) and WBC (0·85) (Fig. 4b). Simultaneously, the cutoff values were calculated from the ROC curves, with a value of 21·9 for N8R (Specificity: 92·6%, Sensitivity: 84·6%), 5·0 for NLR (96·3%, 84·6%), 3·2 for NE (81·5%, 84·6%) and 4·3 for WBC (74·1%, 84·6%). The further univariate analysis including the four factors of N8R>21·9, NLR>5·0, NE>3·2 and WBC>4·3 were used to calculate odds ratios (ORs). The results were obtained for NLR (OR: 143, 95% Cl: 11·72–1745·3), N8R (OR: 68·75, 95% Cl: 8·55–552·68), NE (OR: 22, 95% Cl: 3·646–132·735) and WBC (OR: 55, 95% Cl: 6·779–446·23) within our patient cohort were selected as predictive factors for severe COVID-19. We also performed a 5-fold cross-validation in all patients by using univariate logistic regression for each variable. The average of the AUC and the standard deviation are listed below: N8R: 0·908 ± 0·145, NLR: 0·935 ± 0·093, WBCC: 0·868 ± 0·141, NEC: 0·828 ± 0·198. The predictive performances of N8R and NLR were still significant when fungal co-infection cases were excluded from the group (Fig. S2b).

4. Discussion

In this study, we analyzed the clinical features and immunological characteristics of peripheral blood in patients with COVID-19. Although the majority of the patients did not have an exposure history to the Huanan seafood market in Wuhan, the clinical characteristics of these patients are very similar to those reported in previous studies [3,5,11]. In severe patients, the ages as well as the proportion of underlying diseases are higher, and co-infection also occurs. Recent reports show that the lymphocyte counts are normal in COVID-19 patients with mild diseases. In contrast, 63%–70·3% of patients with severe diseases have lymphopenia and the lymphocyte

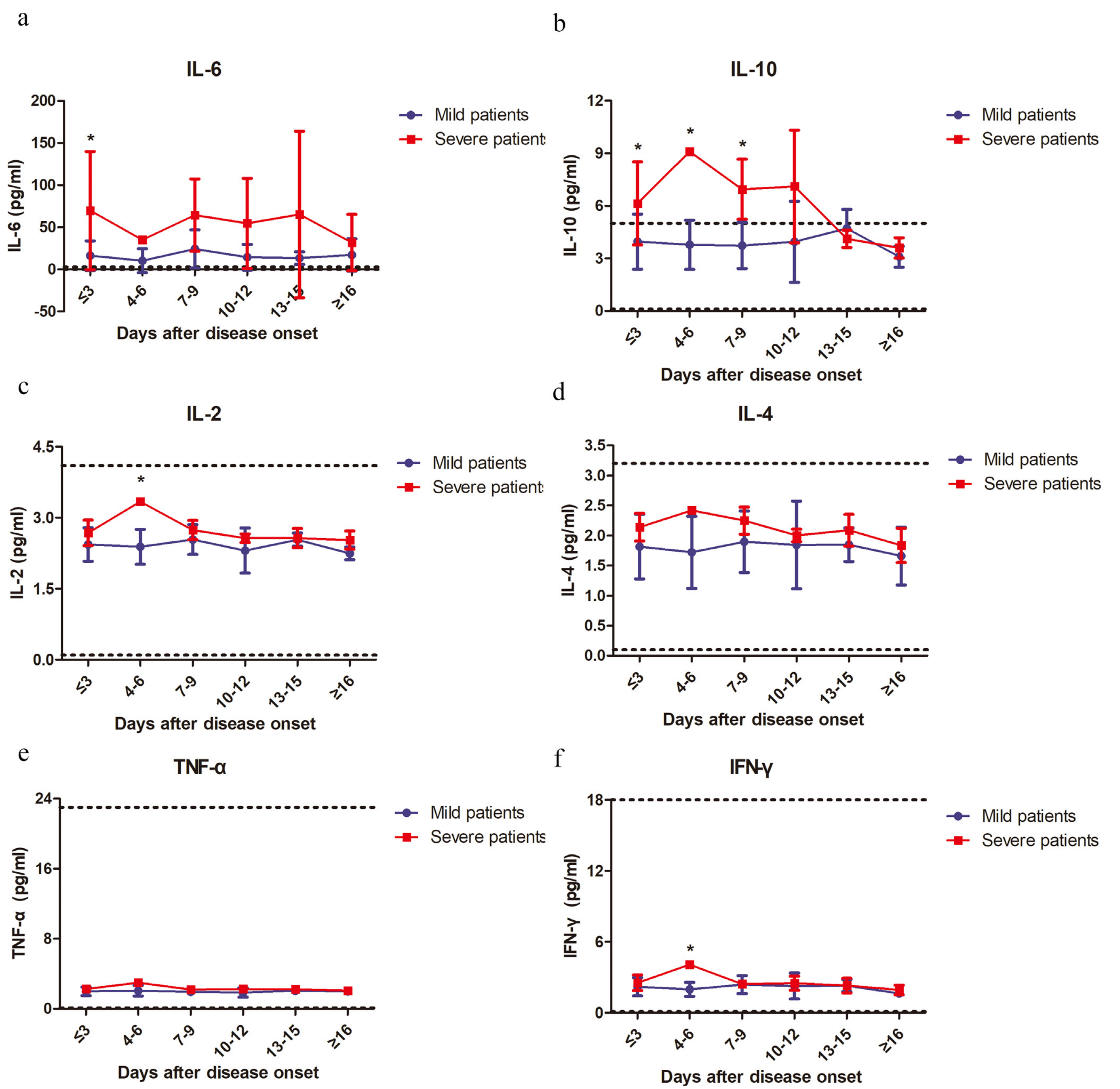

Fig. 3. Kinetic analysis of the serum levels of inflammatory cytokines in COVID-19 patients. The concentrations of IL-6 (a), IL-10 (b), IL-2 (c), IL-4 (d), TNF-α (e) and IFN-γ (f) in the serum of mild (blue line) and severe (red line) COVID-19 patients were analyzed at different time points after hospital admission. Error bars, mean ± SD.; Statistics, repeated measures (mixed model) ANOVA. $^*p<0{\cdot}05$. The upper dotted lines show the upper normal limit of each parameter, and the lower dotted lines show the lower normal limit of each parameter. Number of cases for each time point, M: Mild patients, S: Severe patients. Time≤3d 27(M),13(S); 4-6d 10(M), 1(S); 7-9d 9 (M), 4 (S); 10-12d 6 (M), 3(S); 13-15d 4 (M), 6 (S); ≥16d 2 (M), 7 (S).

counts in patients with a mortal outcome remain at a low level [5,11]. Our study also confirmed higher rates of developing lymphopenia in severe patients than in mild patients (84·6% vs. 44·4%). Recent studies reported that the SARS-CoV-2 infection may primarily affect T lymphocytes particularly CD4$^+$ T and CD8$^+$ T cells, which might be highly involved in the pathological process of COVID-19 [10]. We also found that the development of lymphopenia in severe patients was mainly related to the significantly decreased absolute counts of T cells, especially CD8$^+$ T cells, but not to the absolute counts of B cells and NK cells. The decrease of T cells in the severe patient group reached its peak within the first week during the disease course, and then T cell numbers gradually increased during the second week and recovered to a comparable level to that of the mild patient group in the third week. All the severe patients included in our study survived the disease, and thus we speculate this course is associated with a favorable outcome in severe COVID-19 patients.

Previous studies have shown that elevated levels of proinflammatory cytokines, such as IFN-γ, TNF-a, IL-6 and IL-8, are associated with severe lung injury and adverse outcomes of SARS-CoV or MERS-CoV infection [7,12–14]. Our results also demonstrate that severe COVID-19 patients have higher concentrations of IL6, IL10, IL2 and IFN-γ in the serum than mild cases, suggesting that the magnitude of cytokine storm is associated with the disease severity. Additionally, T cells are important for dampening overactive innate immune responses during viral infection [15,16]. Thus, a loss of T cells during SARS-CoV-2 infection may result in aggravated inflammatory responses, while a

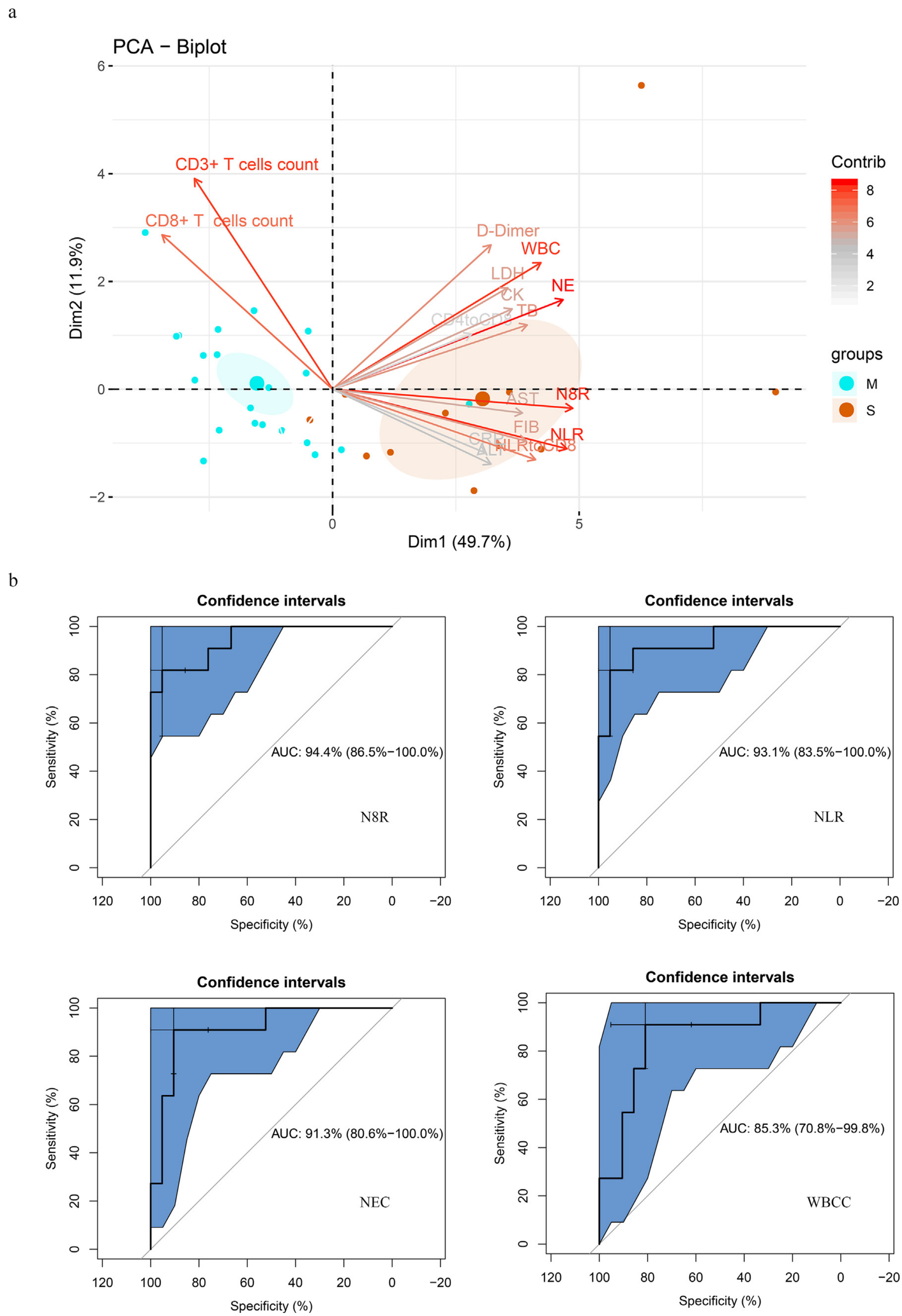

Fig. 4. Prognostic factors of severe COVID-19. (a) Principal component analysis was performed by R package "factoextra" to identify correlated variables for distinguishing severe patients from mild COVID-19 patients. Four mostly contributing variables, neutrophil-to-CD8^{+} T cell ratio (N8R), neutrophil-to-lymphocyte ratio (NLR), neutrophil counts (NE) and White Blood Cells counts (WBC) were identified. (b) ROC curve and AUC were calculated for these 4 selected parameters by using R package "pROC". The results of this analysis identified N8R with a higher AUC (0·94) than NLR (0·93), NE (0·91) and WBC (0·85).

restoration of T cell numbers may alleviate them. In line with this hypothesis, we observed that the kinetic changes of T cell counts are reversely correlated with the kinetic changes of most examined cytokine levels in the peripheral blood in severe COVID-19 patients. While T cell counts drop to their lowest levels at 4–6 days after disease onset, serum IL-10, IL-2, IL-4, TNF-α and IFN-γ levels reach their peaks. The courses of restoring T cell numbers are associated with decreases of serum IL-6, IL-10, IL-2, IL-4, TNF-α and IFN-γ levels.

The early identification of risk factors for severe COVID-19 patients may facilitate appropriate supportive care and prompt access to the intensive care unit if necessary. A recent study in a 61-patient cohort reported that the NLR was the most useful prognostic factor affecting the prognosis for severe COVID-19 [17], and immunological markers (e.g., $CD4^+$ T cell, $CD8^+$ T cell, NLR) tend to be an independent predictor for COVID-19 severity and treatment efficacy [9,18]. The severity of pathological injury during SARS or MERS correlates with the extensive infiltration of neutrophils in the lung and increased neutrophil numbers in the peripheral blood [19]. Thus, the magnitude of increase in neutrophil counts may suggest the intensity of inflammatory responses in COVID-19 patients. Besides, the magnitude of the decrease in lymphocyte counts also indicates the extent of the impairment of immune system by the viral infection. Therefore, NLR may serve as a useful factor to reflect the intensity of imbalance of inflammation and immune responses in COVID-19 patients. In this study, we also screened the potential prognostic factors affecting the incidence of severe illness in our patient cohort. Based on our findings with analyzing lymphocyte subsets, we further included the ratio of neutrophils to different lymphocyte subsets as parameters. Our kinetic analysis revealed that $CD8^+$ T cells are the major lymphocyte subset which decreases in cell numbers during COVID-19. In line with this finding, our results demonstrate that N8R has a comparable performance with NLR in the ROC curve analysis, and both N8R and NLR may serve as powerful factors for predicting the severe illness incidence in COVID patients.

In summary, our study of the immunological characteristics of the peripheral blood in COVID-19 patients shows that the numbers of neutrophils and T cells, especially $CD8^+$ T cells, as well as the levels of inflammatory cytokines in the peripheral blood is dynamically correlated with the severity of the disease. To the best of our knowledge, this is the first work to describe the kinetic changes of lymphocyte subsets and cytokine profiles in COVID-19 patients. Importantly, we identified N8R and NLR as powerful prognostic factors for the early identification of severe COVID-19 cases. This work may help to achieve a better understanding of immune function disorder as well as immunopathogenesis during SARS-CoV-2 infection, and help physicians to provide timely intervention for COVID-19.

Declaration of Competing Interest

The authors disclose no conflicts of interest.

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Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ebiom.2020.102763.

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Erratum

Erratum regarding previously published research papers

The following Author Contribution statements were not included in the published versions of the Research Papers that appeared in previous volumes of *EBioMedicine*. The appropriate Author Contribution statements are included below.

Celastrol-induced degradation of FANCD2 sensitises pediatric high-grade gliomas to the DNA-crosslinking agent carboplatin. (*EBioMedicine* **50:** 81–92)

Author contributions: D.S.M. and E.H. conceived and designed the project. D.S.M., M.H.M., P.W. developed and validated the *in vitro* and *in vivo* models used in the study. D.S.M., B.B., M.H.M., and P.W. performed the functional *in vitro* experiments. D.S.M., P.W., and H.M. performed the functional *in vivo* experiments. J.K. provided bioinformatical expertise and support. B.B. and M.H.M., provided material and logistical support and advised on the project. G.J.K. and E.H. acquired funding and supervised the study. All authors contributed to writing the manuscript.

Epigenetically upregulated GEFT-derived invasion and metastasis of rhabdomyosarcoma via epithelial mesenchymal transition promoted by the Rac1/Cdc42-PAK signaling pathway. (*EBioMedicine* **50:** 122–134)

Author contributions: CL and FL designed the whole study and wrote the manuscript. LZ, WC, YP, JD, ZL, QL, HS, LM, WL, YW, YL, PW, YX, YW, LS, JH, and WZ contributed to experimental design and data collection. All authors have agreed with the manuscript and provide their consent for publication.

Combined identification of three miRNAs in serum as effective diagnostic biomarkers for HNSCC. (*EBioMedicine* **50:** 135–143)

Author contributions: CL and QZ conceived the study. ZYY, SYH, and DSZ participated in the study design. QZ and YYJ conducted the study, including acquisition, analysis, and interpretation of data. CL, ZZY, and ZWS drafted the manuscript. All authors critically reviewed, edited, and approved the manuscript and made the decision to submit for publication. All authors assume responsibility for the accuracy and completeness of the data and for the fidelity of the study to the protocol.

Phosphorylated Rasal2 facilitates breast cancer progression. (*EBioMedicine* **50:** 144–55)

Author Contributions: X.W., Y.K. and Z.M.Q. conceived, organized and supervised the study; X.W., M.Y.L. and Y.L.Y.performed the experiments and data collection; Y.L.Y., X.W., C.Q.and K.Y. contributed to the analysis of data and double checking. X.W., C.Q., Y.K., and Z.M.Q. prepared, wrote and revised the manuscript.

Sprouty4 correlates with favorable prognosis in perihilar cholangiocarcinoma by blocking the FGFR-ERK signaling pathway and arresting the cell cycle. (*EBioMedicine* **50:** 166–177)

Author contributions: Q.B, C. TL, S. RQ, L. ZL, Z. XM,and L. ZP carried out experiments. Z.ZL collected the samples. X. YF analysed data. X. YF conceived experiments and wrote the paper. All authors had final approval of the submitted and published versions.

Analysis of gene expression signatures identifies prognostic and functionally distinct ovarian clear cell carcinoma subtypes. (*EBioMedicine* **50:** 203–210)

Author contributions: RYH, TZT, and DSPT, designed and conceptualised the study. DL processed and reviewed OCCC samples. JY performed sample collection and experiments. NYLN curated and reviewed the clinical data of NUH cohort. TZT performed bioinformatics analyses. RYH, TZT, CVY, NYLN and DSPT analysed the data, interpret the results, and wrote the manuscript.

Pro-inflammatory monocyte profile in patients with Major Depressive Disorder and suicide behavior and how ketamine induces anti-inflammatory M2 macrophages by NMDAR and mTOR. (*EBioMedicine* **50:** 290–305)

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Author contributions: W.N. designed and performed *in vitro* experiments, analysed and discussed results, and critically revised the manuscript. L.N.G. and D.E.R. recruited and followed up patients with MDD and performed sample collection. I.G.E. and N.E. performed *in vitro* experiments. A.R.A. processed samples of patients with MDD. M.P.A. and L.M.S. designed and performed *in vivo* murine experiments, analysed and discussed results, and critically revised the manuscript. F.M.D. conceived and designed the study, recruited and followed up patients with MDD, discussed results, and wrote the manuscript. E.A.C.S. and A.E.E. conceived and designed the study, designed and performed experiments, analysed and discussed results, and wrote the manuscript.

Radiomics analysis of placenta on T2WI facilitates prediction of postpartum hemorrhage: A multicentre study. (*EBioMedicine* **50:** 355–365)

Author Contributions: Conception and design: Xiaoan Zhang, Jie Tian, Meiyun Wang. Collection and assembly of data: Qingxia Wu, Kuan Yao, Zhenyu Liu, Longfei Li, Xin Zhao, Shuo Wang, Honglei Shang, Yusong Lin, Zejun Wen. Development of methodology: Kuan Yao, Zhenyu Liu, Longfei Li, Shuo Wang, Yusong Lin, Jie Tian. Data analysis and interpretation: All authors. Manuscript writing: All authors. Final approval of manuscript: All authors.

TP63 Isoform Expression is Linked with Distinct Clinical Outcomes in Cancer. (*EBioMedicine* **51:** 102,561)

Author contributions: A.B. designed experiments, analyzed data and wrote the manuscript. T.M. performed PCR and RT-PCR experiments. Y.W. performed western blot validation experiments. P.B. contributed to statistical design and analysis of data. P.P. supervised experimental design, analyzed data and prepared the manuscript. All authors read and approved of final manuscript.

Serum IGFBP-1 as a potential biomarker for diagnosis of early-stage upper gastrointestinal tumor. (*EBioMedicine* **51:** 102,566)

Author contributions: Y-WX designed the study, searched the literature, performed the experiments, analysed and interpreted the data, did the statistical analysis, and wrote the manuscript. HC designed the study, collected patient samples, performed the experiments, analysed, and interpreted the data. C-QH designed the study, collected patient samples, searched the literature, did the statistical analysis, analysed, and interpreted the data. L-YC collected patient samples, performed the experiments, analysed and interpreted the data. S-HY analysed and interpreted the data. L-SH, and HG collected patient samples and clinical data. L-YC, C-TL, X-YH L-HL and S-LC collected patient samples and clinical data. Z-YW, Y-HP, L-YX, and E-ML conceptualized and designed the study, supervised the project, and revised the paper. All authors vouch for the respective data and analysis, and agreed to publish the manuscript.

Diagnostic accuracy and easy applicability of intestinal auto-antibodies in the wide clinical spectrum of coeliac disease. (*EBioMedicine* **51:** 102,567)

Author contributions: Study concept and design: Luigina De Leo, Tarcisio Not. Acquisition of data: Luigina De Leo, Stefano Martelossi, Grazie Di Leo, Matteo Bramuzzo. Analysis and interpretation of data: Luigina De Leo, Tarcisio Not, Stefano Martelossi, Grazia Di Leo, Matteo Bramuzzo, Vincenzo Villanacci, Chiara Zanchi. Drafting of the manuscript: Tarcisio Not, Luigina De Leo. Critical revision of the manuscript: Alessandro Ventura, Vincenzo Villanacci, Matteo Bramuzzo, Chiara Zanchi. Clinical decisions: Stefano Martelossi, Grazie Di Leo, Matteo Bramuzzo. Histological evaluation of biopsy samples: Vincenzo Villanacci. Intestinal antibodies immunoassays: Luigina De Leo, Michela Pandullo, Petra Riznik

Phage display antibody libraries: Fabiana Ziberna. Statistical analysis: Fabiola Giudici. All authors read and approved the final version of the manuscript.

MEF2C Repressor Variant Deregulation Leads To Cell Cycle Re-Entry and Development of Heart Failure. (*EBioMedicine* **51:** 102,571)

Author contributions: AHMP, ACC designed and performed experiments, analyzed data, and wrote the manuscript. SRC, RRO, AS an MLBV designed and performed experiments. JRMS performed the echocardiography in animals. MFC analyzed data. AG, JLF, GCAR and MML provided human samples. JDM discussed the manuscript. KGF designed experiments, analyzed data, and wrote the manuscript. All authors reviewed and commented on the manuscript.

Developments in Zebrafish Avatars as radiotherapy sensitivity reporters – towards personalized medicine. (*EBioMedicine* **51:** 102,578)

Author contributions: R.F. and M.G.F. conceptualized the research; R.F. and B.C. supervised the research; S.F., B.C., V.P and R.F. performed research, acquisition, analysis and interpretation of data; P.F., R.R-T., N.F. provided primary tumor samples; M.J.C, S.V., J.S., performed calculations and set-up the accelerator, O.P., J.S. for fruitful discussions; R.F. and B.C. wrote the manuscript. S.F., C.G., O.P. and M. G.F. did critical reading and editing of the manuscript.

Multi-cancer V-ATPase molecular signatures: A distinctive balance of subunit *C* isoforms in esophageal carcinoma. (*EBioMedicine* **51:** 102,581)

Author contributions: JCVCS performed most of the experiments and analysis. PNN participated in the analysis and acquisition of data. EPC performed the in silico structural models. ARF and LFRP coordinated the project. JCVCS and ARF wrote the manuscript. JCVCS, PNN, TAS, ARF and LFRP performed study design. TAS and PNN participated in the collection of samples. ALOF and FFF provided specialized scientific and technical support. All authors discussed the results and manuscript text. All authors read and approved the final manuscript.

Heterogeneous nuclear ribonucleoprotein A2/B1 is a negative regulator of human breast cancer metastasis by maintaining the balance of multiple genes and pathways. (*EBioMedicine* **51:** 102,583)

Author Contributions: The authors' work in this study is listed as follows: *In vitro* and *in vivo* assays (YL, HL, FL, LBG, RH, CC and XD); RNA immunoprecipitation (YL); dual-luciferase reporter assay (YL and SL), signal pathways analysis (HL), proteomic analysis (YL), EMT markers test (HL, LBG and RH); real-time PCR (YL, SL, KL, LY, HMT, BBC and XL); and tissue microarray analysis (YL, DHX and XLD). SLS designed and supervised the study. YL and SLS analysed data and wrote manuscripts.

Genetic Risk for Dengue Hemorrhagic Fever and Dengue Fever in Multiple Ancestries. (*EBioMedicine* **51:** 102,584)

Author contributions: GP, ML, KH, IL contributed to the design; ML, SE, LG, GK, AB, IL, LP, CP, IF, RS, ED, FB, YR, PB, JN, LW, DS, SP, GP, AW, CR, LP acquisition of data; GP, ML, AB, LG, GK Interpretation of data; GP, ML, PS, IL drafted the manuscript; IF, LW, DS, SP, GP, AW, AB, ED, LG, GK, ML, RS, KH revised it for critical intellectual content; ML, SE, LG, GK, AB, IL, LP, CP, IF, RS, ED, FB, YR, PB, JN, LW, DS, SP, GP, AW, PS, GK, KH approved the final manuscript; PG, ML, PS, SE, IF, LW, DS, SP, GP, AW, JN, AB, ED, LG, GK, RS, KH agree to be accountable for all aspects of the work.

Cortical haemodynamic response measured by functional near infrared spectroscopy during a verbal fluency task in patients with major depression and borderline personality disorder. (*EBioMedicine* **51:** 102,586)

Author contributions: Syeda F. Husain: Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing-review & editing. Tong-Boon Tang: Supervision, Writing - review & editing. Rongjun Yu: Supervision, Writing - review & editing. Wilson W. Tam: Supervision, Methodology, Writing - review & editing. Bach Tran: Supervision, Writing - review & editing. Travis T. Quek: Participant recruitment, Writing – review & editing. Shi-Hui Hwang: Participant recruitment, Writing – review & editing. Cheryl W. Chang: Participant recruitment, Writing – review & editing. Cyrus S. Ho: Supervision, Writing - review & editing. Roger C. Ho: Conceptualisation, Participant Recruitment, Methodology, Writing – review & editing.

Impact of sitagliptin on endometrial mesenchymal stem-like progenitor cells: A randomised, double-blind placebo-controlled feasibility trial. (*EBioMedicine* **51:** 102,597)

Author contributions: Study concept, design, and overall supervision: J.J.B, S.Q. Prepared manuscript: J.J.B., S.Te., E.S.L., S.Q. Edited manuscript: L.L., L.J.E., M.J.M.D.C., K.J.F., J.M., P.J.B., A.P., P.K.K., R.F. Obtained funding: S.Q., J.J.B, S.Ta. Regulatory approvals: S.Q., S.Te. Patient enrolment, consenting, ultrasound and clinical assessments: S.Q., S.Te., A.P., L.J.E., L.L. CFU assays and analysis: E.S.L., P.J.B. Exploratory investigations: E.S.L., R.F., P.J.B, J.M., K.J.F., M.J.M.D.C., J.J.B. Data analysis: P.K.K., E.S.L., S.Te., J.J.B., S.Q.

The CD24+ Cell Subset Promotes Invasion and Metastasis in Human Osteosarcoma. (*EBioMedicine* **51:** 102,598)

Author contributions: Zhenhua Zhou wrote the manuscript. Zhenhua Zhou, Yan Li and Muyu Kuang performed cell culture, real-time PCR, flow cytometry and animal experiments. Xudong Wang carried out cell migration, invasion, proliferation assays, Western blot and protein mass spectrometry. Jingjing Hu and Jiashi Cao carried out the histological analysis and scores evaluation. Qi Jia and Sujia Wu carried out prognosis statistical analysis of clinical cases. Zhiwei Wang and Jianru Xiao conceived of the study and participated in its designation and helped to draft the manuscript. All authors read and approved the final manuscript.

The Transferability and Evolution of NDM-1 and KPC-2 co-producing Klebsiella pneumoniae from Clinical Settings. (*EBioMedicine* **51:** 102,599)

Author contributions: HW conceived the project and designed the experiments. QW collected samples and performed microbial identification. YL collected the medical records. RW, YL and LJ performed the microbiological experiments. HG performed the computational analyses. YL, HG, RW and HW wrote the manuscript. All authors read and commented on successive drafts and all approved the content of the final version. tumor immune cell infiltration and survival after platinum-based chemotherapy in high-grade serous ovarian cancer subtypes: A gene expression-based computational study. (*EBioMedicine* **51:** 102,602)

Author contributions: RL, WZ and HHZ contributed to the study design. YZ and RH contributed to data collection. RL performed statistical analysis, interpretation and drafted the manuscript. All authors contributed to critical revision of the final manuscript. RL approved the final version of the manuscript.

Mucosal microbial load in Crohn's disease: a potential predictor of response to fecal microbiota transplantation. (*EBioMedicine* **51:** 102,611)

Author contributions: C.M. and G.S. conceived and supervised the study. G.S., E.V., D.C., A.S, J.W. performed the experiments and data analysis. M.P. and C.M. performed the 16S rRNA data analysis and interpretation. S.L., M.M. and E.E. provided the explant tissues and reviewed the manuscript. C.E. provided the patients' clinical data. K.M. and S.V. provided the mucosal biopsies from CD patients and reviewed the manuscript. G.S. and C.M. wrote and reviewed the manuscript. A.C. revised the manuscript. All authors read and approved the final version of the manuscript.

Mesenchymal stem cells ameliorate β cell dysfunction of human type 2 diabetic islets by reversing β cell dedifferentiation. (*EBioMedicine* **51:** 102,615)

Author contributions: Conceptualization, Z.S., S.W.; Funding acquisition, Z.S., S.W.; Study design, L.W., T.L., R.L.; Investigation, L. W., T.L., R.L.; Data analysis, L.W., T.L., R.L.; Methodology, L.W., T.L., G. W., R.L., N.L., B.Z., Y.J.L., X.D., X.C., Y.L.; Data interpretation, S.W., Z.S., Z.W., X.X.; Supervision, S.W., Z.S., C.R.; Writing – original draft, R.L., L. W; Writing – review & editing, Z.S., S.W., X.X., C.R..

A practical model for the identification of congenital cataracts using machine learning. (*EBioMedicine* **51:** 102,621)

Author Contributions: HL, DL, WC, and YL contributed to the concept of the study and critically reviewed the manuscript. HL, DL, JC, ZL, YX, and XL designed the study and performed the literature search. HL, DL, JC, ZL, XL, XW, ZL, and WC collected the data. KZ, JH, LZ, and CG contributed to the design of the statistical analysis plan. DL, KZ, and JLH performed the data analysis and data interpretation. DL and HL drafted the manuscript. HL, DL, CC, YX, LW, and YZ critically revised the manuscript. HL, DL, WC, and YL provided research funding, coordinated the research and oversaw the project. All authors reviewed the manuscript for important intellectual content and approved the final manuscript.

MiR-765 functions as a tumor suppressor and eliminates lipids in clear cell renal cell carcinoma by downregulating PLP2. (*EBioMedicine* **51:** 102,622)

Author contributions: WX, CW and XPZ designed and performed the experiments. WX, JCX and CW wrote the manuscript. WX, KC and TW analyzed and performed the experiments. XGW and XPZ directed the experiments and analyzed and assembled the data. All authors read and approved the submitted manuscript.

Breast cancer induces systemic immune changes on cytokine signaling in peripheral blood monocytes and lymphocytes. (*EBioMedicine* **51:** 102,631)

Author contributions: LW and PPL designed experiments; LW, DLS, TYT and CA conducted experiments; LW and XL analyzed experimental data; AYC, FMD, JY, JW identified and recruited patients into this study; LW and PPL wrote manuscript. All authors read and approved the manuscript.

Near Infrared Photoimmunotherapy Targeting DLL3 For Small Cell Lung Cancer. (*EBioMedicine* **51:** 102,632)

Author contributions: The all authors checked and approved the final version of the manuscript. Y.I. and K.S. mainly conducted all experiments, performed analysis and wrote the manuscript; K.T., S.T., H.Y., Y.N., R.E., M.S., C.K., N.K., H.Y., Y.B., and Y.H. conducted analysis; S.N., T.F., K.K. and T.F.C.Y. conducted surgical operation to gather the specimens; K.S. supervised and conducted the project.

Gut microbiota composition during infancy and subsequent behavioural outcomes. (*EBioMedicine* **51:** 102,640)

Author contributions: AL and PV proposed the analysis. AL, MOH, ALP, and FC contributed to the statistical analysis. AL, PV, ALP, MOH, CS, FC, and MT contributed to data interpretation. FC contributed to biobanking. AL, PV, and MOH drafted the manuscript. All authors provided feedback and edits to the manuscript. Relevant grant funding applications were prepared by and awarded to: PV, ALP, JC, CS, FC, MT, SR, KA, RS, LH, PS, and the BIS Investigator Group.

Intracavernous injection of size-specific stem cell spheroids for neurogenic erectile dysfunction: efficacy and risk versus single cells. (*EBioMedicine* **52:** 102,656)

Author contributions: ZQL and YT designed the whole experiments and guided the entire experiments, and are responsible for the integrity of the data and the accuracy of the data analysis; YDX and ZQL contributed to performance the animal experiments, data analysis and manuscript drafting. HZ, CH, XMZ, YCZ, and RLG contributed to the performance of the experiments. ZCX and ZQL analyzed and interpreted the data, and critically revised the manuscript for important intellectual content. All authors approved the final version of the manuscript.

Identification and external validation of IgA nephropathy patients benefiting from immunosuppression therapy. (*EBioMedicine* **52:** 102,657)

Authors contributions: Research idea and study design: Z-HL, G-TX, C-HZ, T-YC, E-YX, T-GC; data acquisition: Z-HL, C-HZ, T-YC, E-YX, T-GC, XL; data analysis/interpretation: Z-HL, C-HZ, T-YC, E-YX, T-GC, XL, YZ; statistical analysis: T-YC, E-YX,T-GC,YZ; supervision or mentorship: Z-HL, C-HZ, YQ, S-SL, FX, D-DL. All authors read and approved the final version of the manuscript.

Classification of Primary Liver Cancer with Immunosuppression Mechanisms and Correlation with Genomic Alterations. .(*EBioMedicine* **52:** 102,659)

Authors contributions: H.N. conceived the study. M.F., R.Y., T.H, S.H., and K.K. performed the analysis. K.M., K.N., A.F., M.U., S.H., H.A., H.Y., K.C., and S.I. contributed materials and data. K.A. performed immunohistochemical analysis. S.S. and S.T. performed cell line experiments and expression analysis. H.T. and S.M. contributed to the supercomputer environment. M.F. and H.N. wrote the manuscript.

Silencing of circular RNA HIPK2 in neural stem cells enhances functional recovery following ischaemic stroke. (*EBioMedicine* **52:** 102,660)

Author contributions: H.Y. conceived and supervised this project. H.Y and G.W. designed the experiments. G.W., B.H., L.S., S.W., L.Y., J.L., F.W., M.L., S.L., F.Z., Y.Z., Y.B., Y.M. and B.C. conducted experiments and acquired, analysed and interpreted the data. H.Y. and G.W. wrote the manuscript. All authors read and approved the final version of the manuscript.

Genome-wide identification of FHL1 as a powerful prognostic candidate and potential therapeutic target in acute myeloid leukaemia. (*EBioMedicine* **52:** 102,664)

Author contributions: CC and YF designed the study. YF, MX, ZC performed the experiments. YF, ZY, ZZ, XY and XH analyzed the data. CC, MZ and XW obtained the funding. YF, MX, MZ and XW prepared the figures. YF, MX, ZC and CC wrote the manuscript. CC supervised the study. All authors read and approved the final manuscript.

Longitudinal Serum Autoantibody Repertoire Profiling Identifies Surgery-Associated Biomarkers in Lung Adenocarcinoma. (*EBioMedicine* **52:** 102,674)

Author contributions: S-C. T. and H-C. L. developed the conceptual ideas and designed the study. Y. L, S-J. G., H-W. J. performed the experiments, C-Q. L. and W. G. collected the sera samples. S-C.T., H-C. L., Y. L. and C-Q. L. wrote the manuscript with suggestions from the other authors.

A comprehensive analysis of candidate genes in familial pancreatic cancer families reveals a high frequency of potentially pathogenic germline variants. (*EBioMedicine* **53:** 102,675)

Author contributions: Study design: JE, NM and AC. Data collection: JE, MEC, VP, RF, MRG, TRA, LRD, ICG, MR, EMC and MM. Experimental work: JE, CG, JE2, EB, SGM, DG, GM. Data Analysis: JE, JE2, EB, DG, GM and JR. Interpretation of the data: JE, VP, RF, TRA, LRD, ICG, MR, EMC, NM and AC. Preparation of the manuscript: all authors

CircRNA-CIDN mitigated compression loading-induced damage in human nucleus pulposus cells via miR-34a-5p/SIRT1 axis. (*EBioMedicine* **53:** 102,679)

Author Contributions: Q.X. and L.K. designed the study protocol and wrote the manuscript; Q.X., L.K. and J.W. conducted the experiments; Z.L. and Y.S. established the ex vivo IVD cultured model; K.Z. and K.W. collected and analysed data; C.Y. collected the NP tissues and supervised the study; Y.Z. supported and supervised the study.

FGFR1 and FGFR4 oncogenicity depends on N-Cadherin and their co-expression may predict FGFR-targeted therapy efficacy. (*EBioMedicine* **53:** 102,683)

Author contributions: Conceptualization: A.Q., I.F., S.M.P., A.C., and L.P.A.; Methodology: A.Q., A.C., S.V.C, I.F., and S.M.P.; Investigation: A.Q., A.C., I.F., S.V.C., L.P.A. and S.M.P.; Validation: A.Q.,A.M., L.O., E.G., S.V.C, S.M.G, L.M, S.G. and F.L.R.; Formal Analysis: A.Q., I.F., J.Z., S. M.P; Writing – Original Draft: A.Q., I.F., A.C., S.M.P. and L.P.A., Writing – Review & Editing: A.Q., I.F., S.V.C, A.C., S.M.P. and L.P.A., Supervision: A.C., I.F., S.M.P., and L.P.A.; Funding Acquisition: S.M.P., I.F. and L.P.A. All authors read and approved the final version of the manuscript.

BAP18 is involved in upregulation of CCND1/2 transcription to promote cell growth in oral squamous cell carcinoma. (*EBioMedicine* **53:** 102,685)

Author contributions: Xue Wang, Chunyu Wang, and Guangqi Yan designed the study and wrote the manuscript, Xue Wang, Ge Sun, Yuanyuan Kang, Shengli Wang, Renlong Zou, Hongmiao Sun and Kai Zeng performed experiments and analyzed the data, Huijuan Song, Wei Liu, Ning Sun, and Wensu Liu conducted bioinformatic analyses and statistical analyses, Yue Zhao wrote and revised manuscript. All authors read the approved the final manuscript.

Systematic identification of CDC34 that functions to stabilize EGFR and promote lung carcinogenesis. (*EBioMedicine* **53:** 102,689)

Author Contributions: The project was conceived and designed by G.B.Z. The experiments were conducted by X.C.Z, G.Z.W., Q.H., L. W.Q., S.H.G., J.L., L.M., Y.F.Z., C.Z., H.Y., D.L.Z., and M.W.. Biospecimens were harvested/provided by Z.S.W., Y.C.Z., Y.C.H., B.Z., C.L.W., and Z.L.. The EGFR transgenic mice were provided by L.C.. Data were analyzed by G.B.Z., Y.Z., Z.L., L.C., and X.C.Z.. The manuscript was written by G. B.Z.. The study sponsor had no role in the design of the study; the data collection, analysis, or interpretation; the writing of the article; or the decision to submit for publication.

CBX4 transcriptionally suppresses KLF6 via interaction with HDAC1 to exert oncogenic activities in clear cell renal cell carcinoma. (*EBioMedicine* **53:** 102,692)

Author contributions: Conception and design of the study: Jiang N, Zhang CZ, Shen HM; Generation, collection, assembly, analysis of data: Jiang N, Niu G, Pan YH, Pan WW, Zhang MF; Drafting and revision of the manuscript: Jiang N, Zhang CZ, Shen HM; Approval of the final version of the manuscript: all authors.

Enhanced O-linked GlcNAcylation in Crohn's disease promotes intestinal inflammation. (*EBioMedicine* **53:** 102,693)

Author contributions: Q.H.S. wrote the manuscript. Z.X.X. contributed to the conception and writing. W.Y.S., Y.L.L., and Z.X.X. designed research; Q.H.S., Y.P.J., M.D.L., D.Z., R.X.Z., J.C., and Y.L., performed research; C.S.Q., Y.S.W., G.L., H.L.Z., Q.D., J.L., Y.L.L., and Z.X.X. analyzed the data. Q.H.S., G.L., H.L.Z., Q.D., and Z.X.X. revised the manuscript. All authors read and approved the final manuscript.

Elevated myocardial SORBS2 and the underlying implications in left ventricular noncompaction cardiomyopathy. (*EBioMedicine* **53:** 102,695)

Author contributions: Yingjie Wei. supervised the work; Yingjie Wei, Chunyan Li. designed the experiments with help from Fan Liu, Shenghua Liu, Haizhou Pan, Haiwei Du, Jian Huang, Yuanyuan Xie, Yanfen Li and Ranxu Zhao. Yingjie Wei, Chunyan Li and Fan Liu analyzed the data; Chunyan Li and Yingjie Wei cowrote the manuscript. All authors discussed the results and commented on the manuscript.

Artificial intelligence-assisted prediction of preeclampsia: development and external validation of a nationwide health insurance dataset of the BPJS Kesehatan in Indonesia. (*EBioMedicine* **54:** 102,710)

Author contributions: HS and ECYS developed the concept and design of this study. Dataset access was requested by HS. This author and YWW, and ECYS had full access to all data in the study. HS extracted and processed the data, performed training and validation of machine learning algorithms, conducted the literature search and wrote the draft of the manuscript. HS, YWW, and ECYS independently assessed the eligibility criteria of reviewed studies. YWW and ECYS critically revised the drafted manuscript. HS and ECYS take responsibility for data integrity and the accuracy of the analysis. All authors reviewed the final manuscript.

Plantar temperatures in stance position: A comparative study with healthy volunteers and diabetes patients diagnosed with sensoric neuropathy. (*EBioMedicine* **54:** 102,712)

Author Contributions: UN, MS, JM, AM and PRM contributed equally to this study. PRM and SK conceived and designed the study. ED, JK, SK, JM, AM, and IW recruited participants and performed the experiments. UN, MS, JM, AM and PRM analyzed the data. UN, MS, JM, and PRM drafted the manuscript. TS and PRM were responsible for the design and performance of the sensor-equipped insoles and for data retrieval.

TRAF4 acts as a fate checkpoint to regulate the adipogenic differentiation of MSCs by activating PKM2. (*EBioMedicine* **54:** 102,722)

Author contributions: SC, JL, ZC and YP designed the study and performed the experiments. ZS, ZL and GY performed the statistical analyses. GZ, ML, WL, WY and SW contributed study material and reagents. SC, ZX, PW and HS wrote the manuscript. ZX, PW and HS are the corresponding authors. All authors read and approved the final manuscript.

Identification, clinical manifestation and structural mechanisms of mutations in AMPK associated cardiac glycogen storage disease. (*EBioMedicine* **54:** 102,723)

Author Contributions: Dan.H, and Dong.H. designed the study. Dong.H., H.B.M, L.W.L., N.B.S., Y.L., B.W., F.Z., B.L.S., A.A., L.M., Y.X., S. W., C.A., M.H.G., P.M.E., Dan.H performed clinical and pathological phenotyping of study subjects. Dan.H, H.B.M, and Dong.H. coordinated the clinical evaluations. Dan.H, H.B.M, M.H.G., P.M.E., and Dong. H. supervised and coordinated the genetic laboratory work. Y.L., Y.X., S.W., Dan.H, and D.B., performed history analysis. H.M., K.M., K.L., Dan.H, and D.B., performed computational modeling calculations and transfer entropy analysis. Dan.H, H.B.M, and Dong.H. organized and summarized the database. Dan.H, H.B.M, L.W.L., D,B. and Dong.H. analyzed the data. Dan.H, D,B. C.A., M.H.G., P.M.E., and Dong.H. developed the conceptual approaches to data analysis. Dan.H, Dong.H. D,B. and H.B.M, wrote the manuscript. All co-authors contributed to critical editing of manuscript.

Precise pulmonary scanning and reducing medical radiation exposure by developing a clinically applicable intelligent CT system: Toward improving patient care. (*EBioMedicine* **54:** 102,724)

Author contributions: Conceptualization: Yang Wang and Bing Zhang; Experimental and data studies: Yang Wang, Xiaofan Lu, Yingwei Zhang, Xin Zhang, Kun Wang, Jiani Liu, and Xin Li; Technical Support: Renfang Hu, Xiaolin Meng, Shidan Dou, Huayin Hao, Xiaofen Zhao, Wei Hu, Cheng Li, and Yaozong Gao; Statistical analysis: Xiaofan Lu and Fangrong Yan; Construction of artificial intelligence network: Renfang Hu, Xiaolin Meng, Shidan Dou, Huayin Hao, Xiaofen Zhao, Wei Hu, Cheng Li, and Yaozong Gao; Manuscript editing: Yang Wang, Xiaofan Lu, Zhishun Wang, Guangming Lu, Fangrong Yan, and Bing Zhang; Funding acquistion: Fangrong Yan and Bing Zhang; Resources: Fangrong Yan and Bing Zhang; Supervision: Fangrong Yan and Bing Zhang. All authors read and approved the final version of the manuscript.

Clinical and genomic insights into circulating tumor DNA-based alterations across the spectrum of metastatic hormone-sensitive and castrate-resistant prostate cancer. (*EBioMedicine* **54:** 102,728)

Author Contributions: Conception of idea, MK; Acquisition of data, MK, WT, LH, KM, HF, EK, AA, SY; Data generation, AW, CM, CW; Analysis and interpretation of data, TZ, JY, MK, AW, CM, CW, PD, HF, EK, AA; Drafting of the manuscript, MK, AA, TZ, JY, WT; Critical revision of the manuscript for important intellectual data, WT, LH, SJ, KM, JY, TZ, SJ, HF, SY, EK, AA; Obtaining funding, MK, LH, AA, EK, KM.

Lifetime risk of autosomal recessive mitochondrial disorders calculated from genetic databases. (*EBioMedicine* **54:** 102,730)

Author contributions: MW and TK conceived the study. JT and MW defined a comprehensive list of mitochondrial disease genes and set up a list of pathogenic variants in these genes, supported by SLS, TMS, and SBW. JT and MW queried two databases (gnomAD and in house) to assess the allele frequencies of disease-causing variants in the general population and calculated the lifetime risks, supported by HP, TM, KO and TK. JT and MW drafted the manuscript which was then refined by all other authors and finalized by MW and TK.

Transcriptional and clonal characterization of B cell plasmablast diversity following primary and secondary natural DENV infection. (*EBioMedicine* **54:** 102,733)

Author contributions: A.T.W conceived of the project, designed and executed experiments, analyzed data, and wrote the paper. G.G. and W.R. designed and executed experiments, analyzed data, and provided subject matter expertise. M.K.M. and B.G. analyzed data. T.L., H.S., K.V., C.K., A.G., M.E.F., and J.L. generated data. A.M, A.S., E.D., S.F. provided subject matter expertise and supervised data generation. B.J.D. secured funding. T.E., S.T., and A.L.R. secured funding and provided subject matter expertise. R.G.J. provided project oversight, secured funding, and provided subject matter expertise. D.E. provided project oversight and subject matter expertise. J.R.C and H.F. conceived of the project, designed and executed experiments and analyzed data.

Zika Virus Envelope Nanoparticle Antibodies Protect Mice without Risk of Disease Enhancement. (*EBioMedicine* **54:** 102,738)

Author contributions: Literature search: SS; Figures: RS, RKS, SS, NK; Study design: SS, NK, JKL, FK; Data collection: RS, RKS, VR, UA, GB, JAA; Data analysis and interpretation: SS, NK, JKL, FK; Writing: SS and NK; Approval of final manuscript: all authors.

Bio responsive self-assembly of Au-miRNAs for targeted cancer theranostics. (*EBioMedicine* **54:** 102,740)

Author contributions: The authors' responsibilities were as follows: WC, LY, YW and XW devised the experiments and wrote the manuscript. WC conducted the synthesis of materials, purification, and materials/biological characterizations etc. HF contributed to the mouse model experiment. All other authors contributed to materials synthesis, purification/characterization, and/or discussion of the results.

Large-scale network dysfunction in the acute state compared to the remitted state of bipolar disorder: A meta-analysis of resting-state functional connectivity. (*EBioMedicine* **54:** 102,742)

Author Contributions: Yanlin Wang and Xiaoqi Huang designed the study, Yanlin Wang and Shi Tang collected data and performed analyses; Lu Lu, Lianqing Zhang, Xinyu Hu, Xuan Bu, Hailong Li, Xiaoxiao Hu, Xinyu Hu, Ping Jiang, and Zhiyun Jia provided helpful suggestions; Yanlin Wang, Yingxue Gao and Shi Tang drafted the main article; John A. Sweeney, Qiyong Gong and Xiaoqi Huang critically reviewed the manuscript.

Dynamics of within-host Mycobacterium tuberculosis diversity and heteroresistance during treatment. (*EBioMedicine* **55:** 102,747)

Author contributions: Study design: CN, JB, FB; Data collection: CN, KB, JM, AG, NP, MO; Data analysis: CN, FB; Data interpretation: CN, JM, MO, FB; Writing: CN, FB; Review and approval of manuscript: CN, KB, JM, AG, NP, MO, JB, FB; All authors have read and approved the final version of this manuscript.

Host transcriptomic signature as alternative test-of-cure in visceral leishmaniasis patients coinfected with HIV. (*EBioMedicine* **55:** 102,748)

Author contributions: All authors read and approved the final version of the manuscript. Wim Adriaensen: Conceptualization, data curation, formal analysis, investigation, visualization, writing & editing Bart Cuypers: Formal analysis, methodology, writing, review & editing Carlota F. Cordero: Formal analysis Bewketu Mengasha: Data collection and curation Séverine Blesson: Data curation, project coordination Lieselotte Cnops: Formal analysis, writing, review & editing Paul M. Kaye: Methodology, supervision, review & editing Fabiana Alves: Data curation, funding acquisition, project administration, review & editing Ermias Diro: Data curation, project coordination, funding acquisition, review & editing Johan van Griensven: Conceptualization, methodology, funding acquisition, project administration, supervision, review & editing

Motor transmission defects with sex differences in a new mouse model of mild spinal muscular atrophy. (*EBioMedicine* **55:** 102,750)

Author Contributions: Marc-Olivier Deguise: Generated the mouse model, designed study, produced and analyzed data for all figures, and wrote the manuscript. Yves De Repentigny: Data acquisition, data analysis and method description. Alexandra Tierney: Data acquisition and data analysis

Ariane Beauvais: Assistance with experiments. Jean Michaud: Assessment of histology of the skeletal muscle. Lucia Chehade: Data acquisition and data analysis. Mohamed Thabet: Assistance with electrophysiology. Brittany Paul: Data acquisition and data analysis. Aoife

Reilly: Assistance with experiments. Sabrina Gagnon: Maintenance of mouse models and genotyping. Jean-Marc Renaud: Electrophysiology and data analysis. Rashmi Kothary: Designed study and wrote manuscript.

Ileo-colonic delivery of conjugated bile acids improves glucose homeostasis via colonic GLP-1-producing enteroendocrine cells in human obesity and diabetes. (*EBioMedicine* **55:** 102,759)

Author Contributions: Conceptualization, AA, MC, FMG, and AV; Methodology, AM, AA, JR, BG, MC, FMG, and AV; Formal Analysis, GC, AM, JR, AA, FMG, and AV; Investigation, GC, AM, AA, JR, JD, IZ, GF, DB, GR, BG, SN, AA. Resources, FR, BG, AV, NFL, FMG, MC, AA. Writing – Original Draft: GC, AM, JR. Writing – Review & Editing, GC, AM, AA, JR, JD, GF, DB, GR, FR, BG, AV, NFL, FMG, MC, AA. Visualization, GC, AM, JR, Supervision FR, BG, AV, NFL, FMG, MC, AA Funding Acquisition, FMG, AA.

Longitudinal characteristics of lymphocyte responses and cytokine profiles in the peripheral blood of SARS-CoV-2 infected patients. (*EBioMedicine* **55:** 102,763)

Authors contributions: Conceptualization: JL, SML, JL, YH, DLY, XZ. Acquisition of data: BYL, XBW, HW, WL, QXT, JHY, LZ, LJX, CXG, JT, JZL, JHY, RP, HS, CP, TL, QZ, JW, LX, SHL, BJW, ZHW, CRH, HBZ, RZ, HLZ, XC, PY, BZ, LW,WQZ SSH,YWH, SHJ, PW, JAZ, YPL, WXW, LZ, LL, FQZ. Analysis and interpretation of data: JL, SML, JW. Writing-original draft Preparation: JL. Writing-review and editing: UD, MJL, JL, DLY, XZ. All authors reviewed and approved the final version of the manuscript.

A dysregulated bile acid-gut microbiota axis contributes to obesity susceptibility. (*EBioMedicine* **55:** 102,766)

Author contributions: Wei Jia was principal investigator of this study. Zhaoxiang Bian provided valuable support for C. scindens gavage animal experiment. Wei Jia, Aaihua Zhao, Xiaojiao Zheng, and Guoxiang Xie designed the study. Meilin Wei conducted key experiments of the study and perform the data analysis and drafted the manuscript. Fengjie Huang, Yunjing Zhang, Wei Yang, and Ling Zhao conducted the animal experiments. Kun Ge, Chun Qu, Mengci Li, Shouli Wang, and Xiaolong Han helped to perform the experiments and collected the data. Wei Jia and Cynthia Rajani revised the manuscript.

Prognostic and predictive value of a five-molecule panel in resected pancreatic ductal adenocarcinoma: A multicentre study. (*EBioMedicine* **55:** 102,767)

Author Contributions: Conception and design: JCG, SL, TPZ. Provision of study material and patients: JCG, SL, TPZ, ZGZ, BS, QL, MHD. Financial and administrative support: JCG, SL. Data analysis and interpretation: PZ, LZ, LY, QFL, ZYL, JL, DY, ADT, JS. Experimental support: PZ, LZ, LY, GGX. Manuscript writing: PZ, LZ, QFL. Final approval of the manuscript: All the authors.

CD24-targeted intraoperative fluorescence image-guided surgery leads to improved cytoreduction of ovarian cancer in a preclinical orthotopic surgical model. (*EBioMedicine* **56:** 102,783)

Author contributions: Literature search: E. McCormack, L. Bjørge, K. Kleinmanns, V. Fosse; Study design: E. McCormack, L. Bjørge, K. Kleinmanns, V. Fosse;

Development of methodology: E. McCormack, L. Bjørge, K. Kleinmanns, V. Fosse; Data collection (*in vitro* data, animal experiments, patient data): K. Kleinmanns, V. Fosse, B. Davidson, O. Tenstad, E. García de Jalón; Data analysis and interpretation of data (statistical analysis): K. Kleinmanns, V. Fosse, B. Davidson, O. Tenstad, E. García de Jalón; Writing, review and/or revision of the manuscript: K. Kleinmanns, V. Fosse, E. McCormack, L. Bjørge; Study supervision: E. McCormack, L. Bjørge. All authors read and approved the final version of the manuscript.

Low oxygen saturation during sleep reduces CD1D and RAB20 expressions that are reversed by CPAP therapy. (*EBioMedicine* **56:** 102,803)

Author contributions: TS, DJG, and SAG conceptualized the association study. TS, RL, RJ, HL, ACG, NK, BEC, JL, and SW performed statistical analysis and data harmonization. All authors critically reviewed the manuscript. YL, JR, and SR collected data and designed components of MESA and its gene expression study. DL collected data and designed components of FOS and the SABRe CVD initiative which collected genes expression data for FOS. RM, SRP, SFQ, SR, and DJG designed and executed the HeartBEAT study, and DJG and AS designed its gene expression study.

Clinical implications of serum neurofilament in newly diagnosed MS patients: A longitudinal multicentre cohort study. (*EBioMedicine* **56:** 102,807)

Author Contributions: FS, VF, TU, M Muthuraman, SGM, SG: Analysis and interpretation of data and drafting the manuscript. AS, RG: Study protocol, design and ethics implementation of the KKNMS cohort study. CL, AS, FL, TK, M Mühlau, LK, TR, A Bayas, A Berthele, FP, HPH, RL, CH, MS, BW, FTB, BT, TK, FW, UZ, UZ, HT, BH, HW, RG: Contributing data and revising the manuscript. SB, FZ: Design and conceptualisation of the study, analysis and interpretation of data, drafting the manuscript.

Molecular analysis of Chinese oesophageal squamous cell carcinoma identifies novel subtypes associated with distinct clinical outcomes. (*EBioMedicine* **57:** 102,831)

Author contributions: Lin Feng and Xiyan Wang designed the study. Meng Liu performed the data collection and data analysis. Wei Sun and Yuan Zhang collected Chinese ESCC samples. Haiyin An and Meng Liu extracted and quantified RNA and DNA. Shujun Cheng provided constructive feedback. Lin Feng and Ruozheng Wang supervised research and provided data interpretation. Meng Liu wrote and reviewed the manuscript.

Using Recombination-Dependent Lethal Mutations to Stabilize Reporter Flaviviruses for Rapid Serodiagnosis and Drug Discovery. (*EBioMedicine* **57:** 102,838)

Author contributions: C.B., X.X, and A.M. performed experiments. K.F. provided critical reagents. C.B., X.X., J.Z., and A.M. analyzed the data. C.B., X.X., J.Z., K.F., and P-Y.S. interpreted results. C.B., X.X., and P-Y.S. wrote the manuscript.

Broadly neutralizing antibodies potently inhibit cell-to-cell transmission of semen leukocyte-derived SHIV162P3. (*EBioMedicine* **57:** 102,842)

Author contributions: Study conception and design: RLG and MC. Acquisition of data: KS, MT, and SH. Management of animals: DD. VL, HM and GS contributed with key reagents and expertise. Analysis and interpretation of the data: KS, NDB, and MC. Draft of the manuscript: KS and MC. Critical revisions: HM, GS, RLG, and MC. All authors read and approved the final version of the manuscript.

GSTM3 variant is a novel genetic modifier in Brugada syndrome, a disease with risk of sudden cardiac death. (*EBioMedicine* **57:** 102,843)

Author Contributions: JMJJ, TPL, and CA performed literature search, conceived and designed the study and the experiments. JMJJ, TPL, AB, IR, SJL, CYJC, LCL, SFSY, EYC, and LPL conducted experiments and analysed the data. JMJJ, JJH, WCC, YBL, LYL, CCY, LTH, and HCH enrolled patients, collected and interpreted data. JMJJ, AB, IR, TPL, and CA wrote the paper.

Tumor budding, poorly differentiated clusters, and T-cell response in colorectal cancer. (*EBioMedicine* **57:** 102,860)

Author contributions: All authors contributed to review and revision. M.G., J.A.N., and S.O.: developed the main concept and designed the study. A.T.C., C.S.F., M.G., and S.O.: wrote grant applications. K.F., J.P.V., J.B., D.J.P., J.A.M., A.T.C., C.S.F., J.K.L., J.A.N., and S.O.: were responsible for collection of tumor tissue, and acquisition of epidemiologic, clinical and tumor tissue data, including histopathological, immunohistochemical, and immunofluorescent characteristics. K.F., J.P.V., J.B., D.J.P., K.H., J.A.M., C.S.F., J.A.N., and S.O.: performed data analysis and interpretation. K.F., J.P.V., J.B., D.J.P., and S.O.:

drafted the manuscript. K.A., K.H., J.K., N.A., T.U., M.C.L., S.G., S.S., M.Z., A.F.L.D.S., T.S.T, H.N., J.A.M., X.Z., K.W., M.G., J.A.N., and S.O.: contributed to editing and critical revision for important intellectual contents.

A surrogate of Roux-en-Y gastric bypass (the enterogastro anastomosis surgery) regulates multiple beta-cell pathways during resolution of diabetes in ob/ob mice. (*EBioMedicine* **58:** 102,895)

Author contributions: F.A., C.A. and C.M. designed the experiments. C.A.; J.C.; C.G.; A.L.; F.M., C.R., J.D.; E.G.; S.M.L., O.T. conducted the experiments. C.A.; F.A.; C.M.; O.T.; C.G. G.R. and R.R. analyzed data. K.C. contributed to patient recruitment and coordinated clinical investigation, patient phenotyping, and sample collection. F.A. and C. A. wrote the manuscript and C.A.; F.A.; C.M.; O.T.; T.S.; C.G.; R.R.; S.L.; R.S.; H.L.S.; E.G. and G.R. contributed to data presentation and the manuscript. All authors reviewed the manuscript. F.A. is the guarantor of this work and, as such, had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Protection Against Mycobacterial Infection: a case-control study of mycobacterial immune responses in pairs of Gambian children with discordant infection status despite matched TB exposure. (*EBioMedicine* **59:** 102,891)

Author contributions: RB and BK conceived and designed the work. RB, MS, AS and UE conducted the clinical recruitment. RB and BS conducted and interpreted the BCG-GFP-LuxFO whole blood assays. BS and MG conducted the in-house interferon gamma release assays. BH conducted and interpreted the cytokine multiplex assays. RB and AK conducted the statistical analyses. RB and BK drafted the work. All authors revised the work for important intellectual content.

Brain Delivery of Supplemental Docosahexaenoic Acid (DHA): A Randomized Placebo-Controlled Clinical Trial. (*EBioMedicine* **59:** 102,883)

Author contributions: IC, NC, BK, DB participated in recruitment and study visits. HNY and MGH did lumbar punctures. XH, NK, and WJM conducted data analysis. NH, NK and MNB did imaging analysis. LD, CM, and HCC planned cognitive testing. AM, AS, BZ assisted with biomarkers. IC, VS, HH, MH, HCC, WJM, MNB, LSS and HNY wrote the manuscript. HNY and LSS designed the study.

Obesity-related hypoxia via miR-128 decreases insulin-receptor expression in human and mouse adipose tissue promoting systemic insulin resistance. (*EBioMedicine* **59:** 102,912)

Author contributions: B.A. and F.L.A. performed experiments *in vitro* and *in vivo*, in mouse systems; B.A. performed human tissue culture studies and analyzed data with the contribution of E.C., M.M., D. M.C., D.P.F and A.B; G.C. and G.N. provided tissues from surgery and clinical information; D.B., V.M. and UK contributed to the analysis of data from mouse experiments; B.A. and E.C. contributed to manuscript draft; F.S.B. helped collecting clinical data and drafted figures; I. D.G. edited the final version of the manuscript and contributed to data interpretation; A.B. conceived and supervised the study and wrote the manuscript.

All authors read and approved the final manuscript.